

"Harrier 3.5 V6: Failed VVT solenoid sets timing over-advanced bank 1 P0011"

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Area: bulletins

1 Condition

Some Harrier 3.5 V6 vehicles may set diagnostic trouble code P0011, indicating camshaft position timing is over-advanced on bank 1. Drivers may report rough idle, hesitation on light acceleration, and a momentary reduction in power with the MIL illuminated. The condition often appears after extended intervals on degraded engine oil. See the Harrier Engine Service Manual :: Engine System Overview for a description of the bank 1 camshaft phaser circuit.

2 Cause

The cause is a stuck-open variable valve timing control solenoid, part number VVT-3360, which allows excessive oil flow to the bank 1 phaser and over-advances cam timing, setting P0011. Internal solenoid spool sticking is commonly the result of sludge or oil contamination restricting free travel. The Harrier Engine Service Manual :: Timing Sub-System describes the phaser oil control logic, and the closely related Falcon Engine Service Manual :: Variable Valve Timing System and Lumen Engine Service Manual :: Valve Timing Sub-System cover equivalent solenoid behavior on sister platforms.

3 Repair Procedure

Replace the VVT control solenoid, part number VVT-3360, following the Harrier Engine Service Manual :: Variable Valve Timing Solenoid Replacement (PROC-VVT-SOL). Perform an engine oil and filter change (PROC-OIL) at the same time, since contaminated oil is the most frequent contributor to solenoid sticking. After installation, clear codes and verify P0011 does not reset under load during a road test. Equivalent solenoid service steps are documented in the Falcon Engine Service Manual :: VVT Solenoid Replacement and the Lumen Engine Service Manual :: Variable Valve Timing Solenoid Service for technicians cross-trained on those platforms.